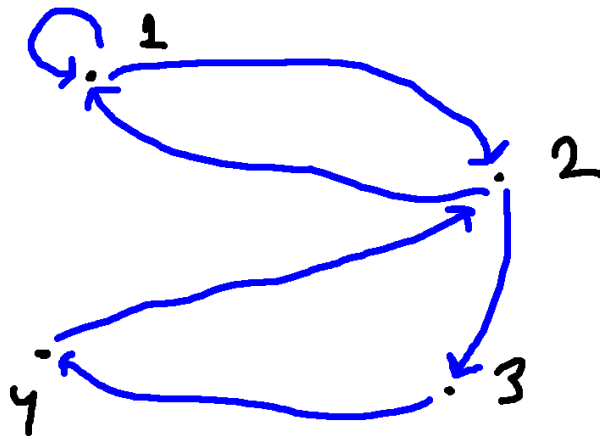


0.1 Section 2.6 - Matrices



Definition 1 (Directed Graph)

This is a **directed graph** and could be used to model:

- How webpages are linked
- Flights between cities
- Who knows who

How can a computer store this directed graph?

Definition 2 (Matrix)

We will use a **matrix**, which is a rectangular array of numbers

For this network, we define a matrix

$$A = [a_{ij}]$$

Where

$$a_{ij} = \begin{cases} 0 & \text{if web page } i \text{ doesn't link to } j \\ 1 & \text{if web page } i \text{ does link to } j \end{cases}$$

i = row (starts at 1)
 j = column (starts at 1)

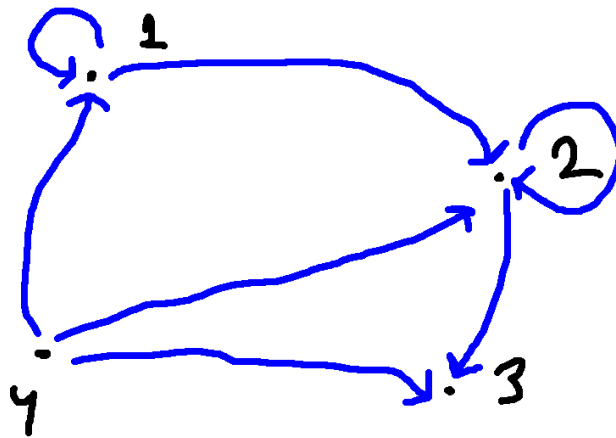
The resulting matrix would look like

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

For this network, the matrix is

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Let's modify the graph now



This matrix is

$$B = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

Matrix Addition and Subtraction

We define the addition of matrices by adding corresponding components

$$A + B = \begin{bmatrix} 2 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

The same can be said for Subtraction

$$A - B = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 1 \\ -1 & 1 & -1 & 1 \end{bmatrix}$$

Note

Two matrices can only be added/subtracted if they're the same size

Question

Going back to the first network, how many ways can we start at 1 and return to 1 after passing through exactly 1 webpage?

$$1 \rightarrow 1 \rightarrow 1$$

$$1 \rightarrow 2 \rightarrow 1$$

$$a_{12} \rightarrow a_{21}$$

$$\begin{bmatrix} a_{11} & a_{12} & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ a_{21} & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Example 1

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

1. $A + B$

Solution:

Undefined, due to dimensions

2. $6A$

Solution:

$$6A = \begin{bmatrix} 6 & 12 & 18 \\ 24 & 30 & 36 \end{bmatrix}$$

3. AB

Solution:

$$AB = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ -1 & 11 \end{bmatrix}$$

4. BA

Solution:

$$BA = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 3 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

5. BC

Solution:

$$BC = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 2 & 4 \end{bmatrix}$$

6. CB

Solution:

Undefined, due to dimensions

7. $C^2 = CC$

$$C^2 = CC = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 10 \\ 10 & 20 \end{bmatrix} \\ = 5C$$

8. C^{11}

$$C^2 = 5C$$

$$C^3 = C^2 \cdot C$$

$$= 5C \cdot C$$

$$= 5C^2$$

$$= 5^2 C$$

$$C^{11} = 5^{10} C = 5^{10} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Okay let's now define some properties

0.1.1 Properties of Matrices

Definition 3 (Additive Identity)

We define the **Additive Identity** for matrices as

$$A_{n \times m} + \vec{0} = A_{n \times m}, \vec{0} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$

Definition 4 (Multiplicative Identity)

We define the **Multiplicative Identity** for matrices as

$${}_{n \times m} A \cdot I_n = A, I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

But A must be **square**!

Some matrices have a multiplicative inverse such that

Definition 5 (Multiplicative Inverse)

We define the **Multiplicative Inverse** of matrices as

$$A \cdot B = I$$

But only **some** matrices have them, not all do!

Also, A must be square to even qualify for possibly having one

For this class, we define

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Example 2

We have

$$\begin{bmatrix} 4 & 3 \\ 7 & 5 \end{bmatrix}^{-1} = \frac{1}{20 - 21} \begin{bmatrix} 5 & -3 \\ -7 & 4 \end{bmatrix}$$